

Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus two additive ones -- an MCP agentic-integration server, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable quantum-explainer PWA). The euro and accuracy figures throughout are on synthetic data (e.g. retail-analytics-real, decision-chain) and are not claims about Wuerth's real business. The real-data projects (retail-analytics-real, decision-chain) are completed: measured on public UCI Online Retail II data (CC BY 4.0) and labelled REAL DATA where cited; decision-chain labels its invented layers on every line.

Job #1 (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth	revops-optimizer, market-basket-analysis	MILP > greedy; 254 rules, top lift 2.41
Pricing & margin	Elasticity + leakage; endogeneity fix	sales-kpi-analytics, ml-models-lab	MASE 0.088, RMSE 0.948 (beats naive + Holt-Winters); policy: EUR127,421 working capital (cycle+safety exactly), 5x returns, exec OK; pacing EUR10.91M = 95.7% of plan, 80% interval -- closed-year back-check landed above the band, reported
Inventory / replenishment	Newsvendor + forecast + ROP/EOQ policy	distributor-intel-platform, ml-models-lab	4.6% / 31% savings; 5% headroom: failing scenarios 96% -> 44%, expected day -4.6%
Sales KPIs & analytics	KPIs + rolling-origin CV + pacing interval	sales-kpi-analytics	
Logistics / routing	OR-Tools CP-SAT VRP + robustness	route-optimizer	
E-procurement automation	Agentic + n8n low-code	agentic-automation-lab	~EUR625k/yr modelled
Document processing	Agentic extract + 7-rule validation gate	doc-extract-agent	~EUR145k/yr modelled; combined-gate precision 74.00% pick 75% level; ABC ROC AUC 0.99, min 0.92400
Customer retention	Decline + churn classifiers	revops-optimizer, ml-models-lab	61.9% 75% 50% 25% slotting businesses; 45% in slotting
Warehouse / WMS (new)	WMS twin + plant sim (Story Mode, 894-element plant, definable objects); slotting + DES + packing pick-path routing	logistics-flow-studio (WarehouseTwin), logistics-digital-twin (engine)	40.32% (golden-zone 25% -> 100%); fill 2.0% -> 30.2%; routing: return +3.0% vs waste optimum 14/14 folds (MASE 0.45% vs Holt-Winters 10.5% to naive/reported); peak; 20.8% 65% EUR180.00/yr at ASSUMED 35 off; battery 750 per server/day (backlog) these assumptions; backtest captures 72.7% of the LP bound (robust variant 2 losses reported)
Supply-network design (new)	Facility MILP + flows + safety stock + service frontier	supply-network-opt	AE 0.17 vs PCA 0.22 margin, PCA recommended; TPR data 7 vs 0.32 @ 5% false call (35% optimization MASE 0.706 (naive 1.02); cost savings 6.5% of top 50 99.04% 864.62 to the penny; faciant to delay; elasticity
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest	energy-demand-forecast	enaholding 25% exactly (0.0580); naive wins lumpy (MASE 1.782); slotting optimum -1.6% vs ABC; CVRP 0.367, swap-set gates -> PROMOTE at 8,632vs8,841 -- retrain finds no new fraud, wins by shedding load (stated) = 44.4 d at 7.16/machine-day -- the calendar rule beats the repo's own detector at the default threshold, reported
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules	quality-anomaly-vision	
Real data (completed)	Cleaning + RFM + returns + leakage-safe CV	retail-analytics-real	
Chain integration & reconciliation (new)	Provenance-tagged pipeline + identity ledger; MCP agentic layer	decision-chain, chain-mcp	
Fraud & transaction-risk ops (new)	Cost-based alerting; champion/challenger gated promotion	fraud-detection-ops	
Maintenance & asset reliability (new)	Censored Weibull + age-replacement vs condition-based	predictive-maintenance	

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Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
Build AI-agent workflows	agentic-automation-lab	agentic tool-use loops
Agentic integration via an open standard (MCP)	chain-mcp (official mcp Python SDK; Claude Desktop / Code configs)	6 real engines as AI-callable tools; typed schemas; never-crash errors; per-tool honesty label; 20 tests incl. live JSON-RPC handshake
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio	flow builder; ~EUR47k/yr
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured
Document intake / understanding	doc-extract-agent + 7-rule validation layer	extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5%
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	runnable prototypes incl. a full installable offline WMS twin + plant simulator (PWA)
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer	hand-written simulator ~300 lines, zero deps; 42 physics assertions + 57 structural checks pass
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless **13/13** reconciliation identities PASS + **14/14** real data)

Requirement	Repo / artifact	Proof (synthetic unless noted)
One dataset through the whole chain (real data + labelled layers)	decision-chain (run artifact + offline dashboard; FAIL path per identity)	== holding share exactly (0.0580); naive wins lumpy (MASE 1.782); slotting proof (syntactic unless noted) vs Clarke-Wright; crew 18% utilized; cost rates labelled INVENTED
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk	DAX pack + 3-page report + Monte-Carlo RPD/BPO/PPQ risk pop-up CI-RX EUR10.91M = 95.7% of plan, 80% interval;
KPI dashboards / metrics	sales-kpi-analytics + pacing bullet chart	back-check: realised EUR11.28M landed above the band, honest miss P10-P90 EUR145,091..170,723/yr, ~43% chance of clearing the headline;
Uncertainty / risk quantification	revops-optimizer simulate.py; ml-models-lab BENCHMARK_CI.md	tornado ranks drivers; bootstrap skill CIs; churn +0.457 [-+0.381, +0.528] ROP/EOD at ABC-XYZ targets: EUR127,421 working capital (cycle+safety exactly),
Inventory policy / working capital	distributor-intelligence-platform dip/inventory.py; supply-network-opt frontier	5.5x turns, 99.9% fill; frontier 6.3x convexity
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE wins MASE 0.987 vs MSECon 0.948 gains renders loss from no-silent Wirt' V=0.02% over EUR15.100y eat AS SUMED
Forecasting rigour + reconciliation guard	distributor-intelligence-platform (MRO command center)	Identify within 2 times baseline EUR; present does not pay back on these assumptions; causal backtest captures 72.7% of the LP bound (robust variant losses)
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP + dispatch backtest; 59 tests)	
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	~EUR160k/yr; -21.2% cost vs greedy; ROE-AUC 0.99; 2% SEP-Bias 0.52 -> +0.03; churn ECE 0.197 -> 0.021, bootstrap skill CI +0.457 [+0.381, +0.528]
Elasticity / statistical modelling	revops-optimizer; ml-models-lab	
Data into business decisions	sales-kpi-analytics; market-basket-analysis	EUR2.6M leakage lever; cross-sell recs, Xpibit 4.4P-growth from scratch; 224 itemsets, 254 rules; affinity network Q=0.58, 3 communities, 4 bridges; 55 tests
Market-basket / cross-sell	market-basket-analysis + affinity.py	SKU macro-F1 0.963; AE PR-AUC 0.963 vs RCA 0.951 (PCA default); fraud PR-AUC 0.70 (PCA) vs 0.94 (DBSCAN) vs 0.95 (ML) vs 0.93 (K-Means) vs 0.92 (Random Forest)
Classification / anomaly detection	ml-models-lab; fraud-detection-ops	WMS twin + grant sim: keyword generated by wordnet, 1316 words, 1000 items, 5000 material 80032 v1.0 (presented)
Model lifecycle / MLOps (gated retrain)	fraud-detection-ops challenger.py	Sims Modeler fraud ops - presented modeling every 24 months (retrained) uses definable object library BYO CSV, predictor models with Keras, 48.6ms inference, 100% accuracy, random 5193740 purchases @ 33% purchased, 100% breaks hard for all post 900 golden, 100% WE 100%, calibration correction 0 <= WE 100%
Reliability / maintenance (survival)	predictive-maintenance pdm/policy.py	DES-76.1% / 66.5%; PERUSH routing: return +3.0% vs exact optimum, optimized layout ~46% shorter
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; SPC; 45 tests)	
Warehouse / intralogistics	logistics-flow-studio (WarehouseTwin); logistics-digital-twin (engine)	4.6% / 31% savings; robustness: 5% headroom cuts failures 96% -> 44%, expected day 0%, low safety stock 0.5M revenue; returns 3.65% of gross, 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; seasonal-naive wins CV, MASE 1.094
Logistics / ops analytics	route-optimizer (36 tests); supply-network-opt (53 tests)	
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	
Present to decision-makers	revops-optimizer exec deck	exec narrative deck